

## Section 5.1

### Eigenvalues and Eigenvectors

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### Eigenvalues and Eigenvectors

An eigenvector of an  $n \times n$  matrix  $A$  is a nonzero vector  $\mathbf{x}$  such that  $A\mathbf{x} = \lambda\mathbf{x}$  for some scalar  $\lambda$ .

A scalar  $\lambda$  is called an eigenvalue of  $A$  if there is a nontrivial solution  $\mathbf{x}$  of  $A\mathbf{x} = \lambda\mathbf{x}$ . Such an  $\mathbf{x}$  is called an eigenvector of  $A$  corresponding to  $\lambda$ .

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### Example 1

Is  $\begin{bmatrix} -1 + \sqrt{2} \\ 1 \end{bmatrix}$  an eigenvector of  $A = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$ ?  
If so, find the corresponding eigenvalue.

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### Example 2

Is  $\lambda = 2$  an eigenvalue of  $A = \begin{bmatrix} 3 & 3 \\ 2 & 8 \end{bmatrix}$ ?

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### Eigenspaces

The equation  $A\mathbf{x} = \lambda\mathbf{x}$  is equivalent to  
 $(A - \lambda I)\mathbf{x} = \mathbf{0}$

The set of all solutions of this equation is the null space of  $(A - \lambda I)$ , and this set is called the eigenspace of  $A$  corresponding to  $\lambda$ .

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### Example 3

Find a basis for the eigenspace of  $A$  corresponding to  $\lambda = 3$ .

$$A = \begin{bmatrix} 4 & 2 & 3 \\ -1 & 1 & -3 \\ 2 & 4 & 9 \end{bmatrix}$$

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### Eigenvalues of a Triangular Matrix

The eigenvalues of a triangular matrix are the entries on its main diagonal.

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### Example 4

Find the eigenvalues of the matrix.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 5 \end{bmatrix}$$

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### Example 5

Construct an example of a  $2 \times 2$  matrix with only one distinct eigenvalue.

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### Example 6

Let  $\lambda$  be an eigenvalue of an invertible matrix  $A$ . Show that  $\lambda^{-1}$  is an eigenvalue of  $A^{-1}$ .

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### The Invertible Matrix Theorem

Let  $A$  be a square  $n \times n$  matrix. Then the following statements are equivalent.

- a.  $A$  is an invertible matrix.
- $\vdots$
- s. The number 0 is not an eigenvalue of  $A$ .

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### Theorem

If  $\mathbf{v}_1, \dots, \mathbf{v}_r$  are eigenvectors that correspond to distinct eigenvalues  $\lambda_1, \dots, \lambda_r$  of an  $n \times n$  matrix  $A$ , then the set  $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$  is linearly independent.

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**Example 7**

Prove the preceding theorem.

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